

Cloud Computing Paradigms

Cloud and Edge Computing
Claudia Canali

Vision and evolution



- ***When and how** was the cloud born?*

Vision and evolution



- [John McCarthy, MIT, 1961]: “**Computing may someday be organized as a public utility**”

The historical 4 utilities

(1) Water



(2) Electricity



(3) Gas



(4) Phone/Internet



Hidden complexity

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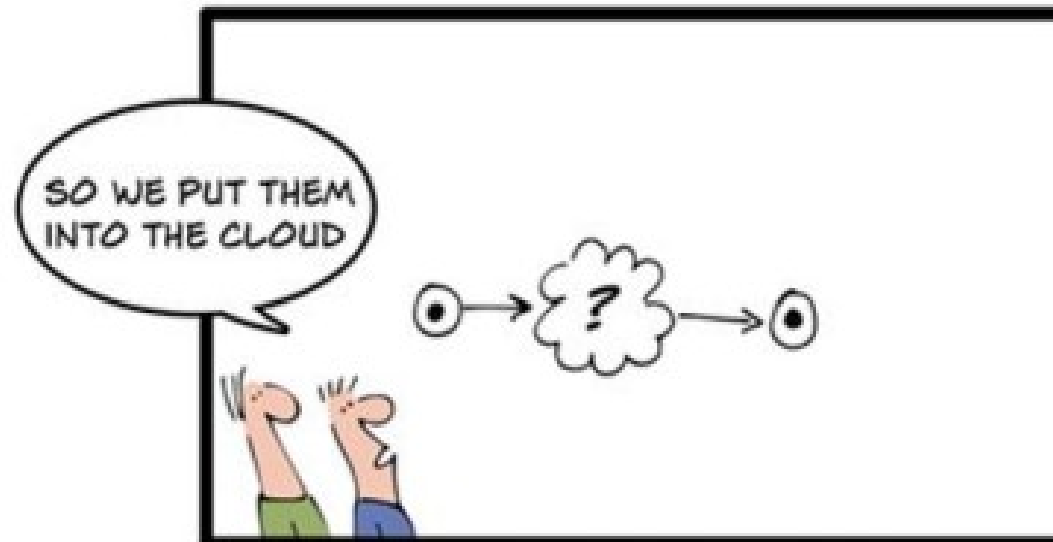
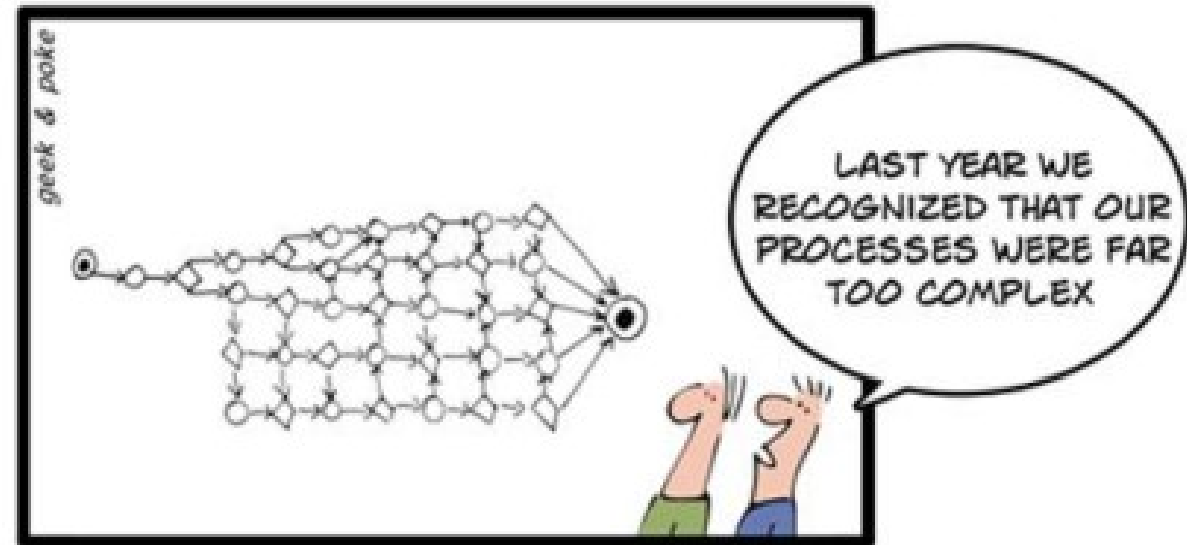


Cloud as...

...evolution of Internet where the company resources (e.g., applications, business processes, collaboration instruments, storage, hw and sw) can be accessed as on online service



Complexity seems disappeared



LET THE CLOUDS MAKE YOUR LIFE EASIER

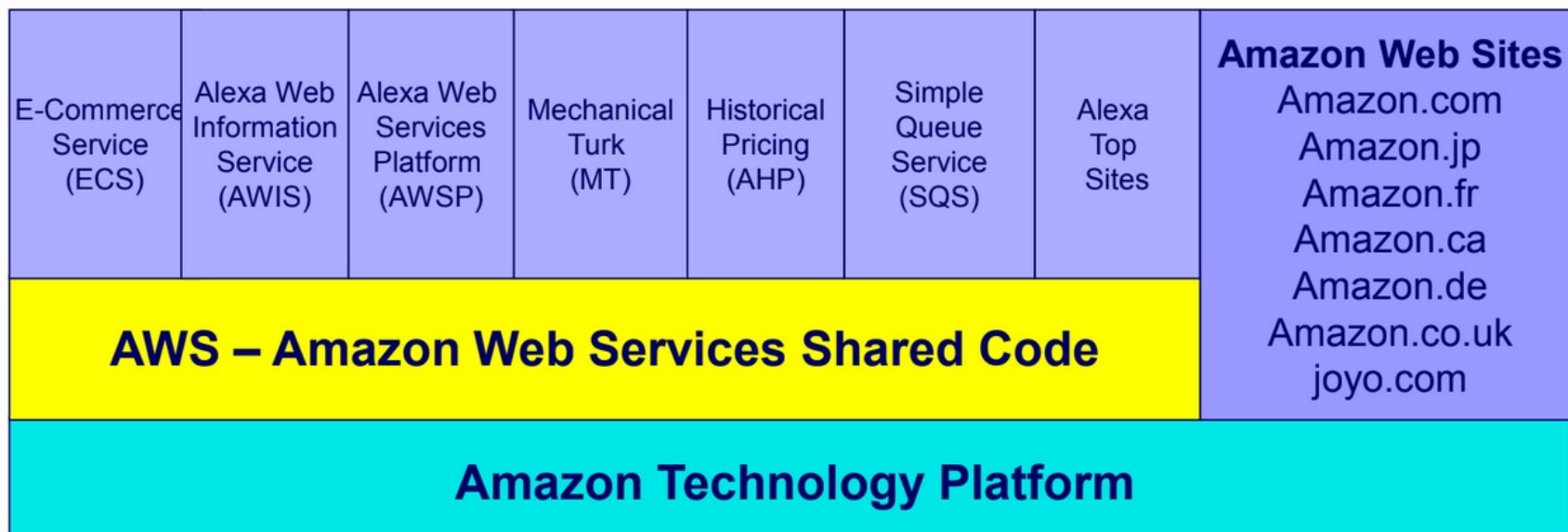
Vision and evolution



- [John McCarthy, MIT, 1961]: **“Computing may someday be organized as a public utility”**
- [Jeff Bezos, CEO Amazon, 2006]: **“Let us use our spare resources for making profit by offering them as services to the public”**
 - The CLOUD COMPUTING was born
- [Steve Ballmer, CEO Microsoft, 2016]: **“The cloud is the future, everything else is accessory”**

Consequence

- [Jeff Bezos, CEO Amazon, 2006]: “It is important to sell unused resources of a datacenter so its cost may become a profit”
- A huge step ahead for Amazon’s engineers:
It means not only that the management of an open system receives an *unpredictable* number of requests from the customers
- **It means sharing the same resources used for internal procedures/business with the customers**
- ***Completely disruptive model of resource sharing***



A first Cloud Computing definition



Many definitions available, not only one

AWS Definition

- *“Cloud Computing is the on-demand delivery of compute power, database, storage, applications and other IT resources via the Internet with pay-as-you-go pricing model”*
- *These resources run on server computer that are located in large data centers in different locations around the world*

The NIST definition

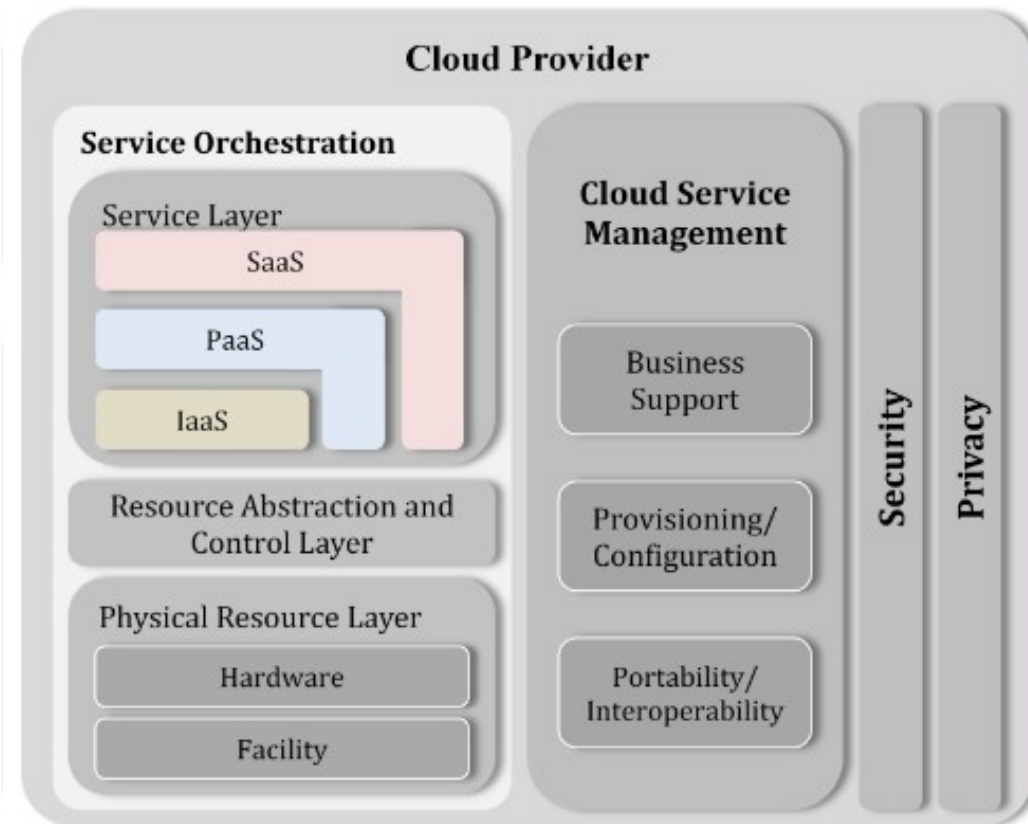


Founded in 1901, the **National Institute of Standards and Technology's (NIST)** is an agency of the United States government that deals with technology management and standards, guaranteeing emerging technologies are secure, reliable and interoperable

- **After 5 years** since cloud was born (from a *private* initiative – differently from other disruptive innovations as Internet and Web) → need for standardization
- **After 15 drafts** → **NIST** published the 16th and final definition of Cloud Computing
- **NIST Cloud Computing Reference Architecture Recommendations of the National Institute of Standards and Technology**
September 2011
https://bigdatawg.nist.gov/_uploadfiles/M0008_v1_7256814129.pdf

The NIST definition

“Cloud computing is a model for enabling **ubiquitous, convenient, on-demand** network access to a **shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned and released with minimal management effort or service provider interaction**. This Cloud model promotes **availability, elasticity, and security**. [...].”



The NIST definition



5 Key Points emerging from the NIST definition

- **On-demand service**
 - No reservation in advances, resource available when needed, no human intermediary in the loops
- **Broad Network Access**
 - Data centers connected at the network with multiple, redundant, high capacity connections
- **Resource pooling**
 - Not infinite but the perception of the customers is almost that – due to its dynamic behavior
- **Rapid elasticity**
 - Related to the on-demand service: speed to have more resource (order of seconds / minutes) and possibility to release resources
- **Measured Service**
 - Capability to measure all the usage of resources done by the customers and monitoring resources

Cloud Paradigms

3 main actors

- **Provider:** provider of cloud services – owner of the cloud infrastructure
- **Customer (organization):** organization that makes use of (and pays for) the cloud services offered by the provider, generally offering an added value as for example the development of software services – usually the *service provider*
- **(Final) User:** user that makes use of the services properly configured and made available by the customer

Two main classes of paradigms



Service

Deployment

Service paradigms (prevalent)



- **Infrastructure as a Service (IaaS)**
- **Platform as a Service (PaaS)**
- **Software as a Service (SaaS)**

Model “as a Service (aaS)”

Building blocks

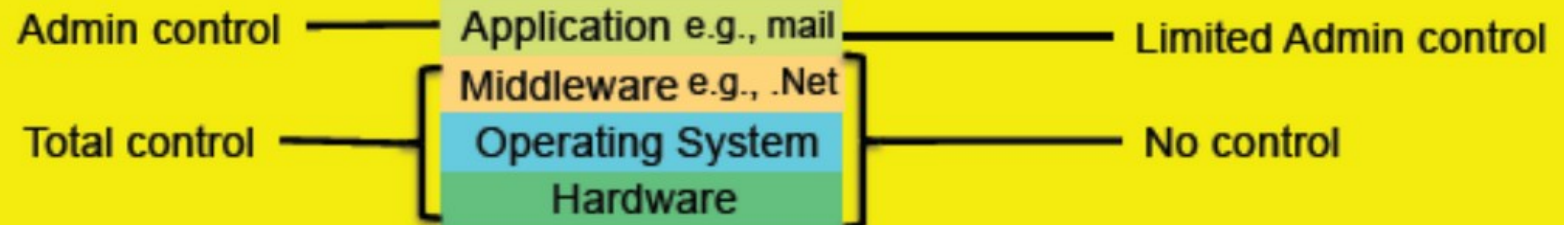
- Shared (multi-tenant)
 - Pay per use
 - Hourly/monthly bill

Service paradigms

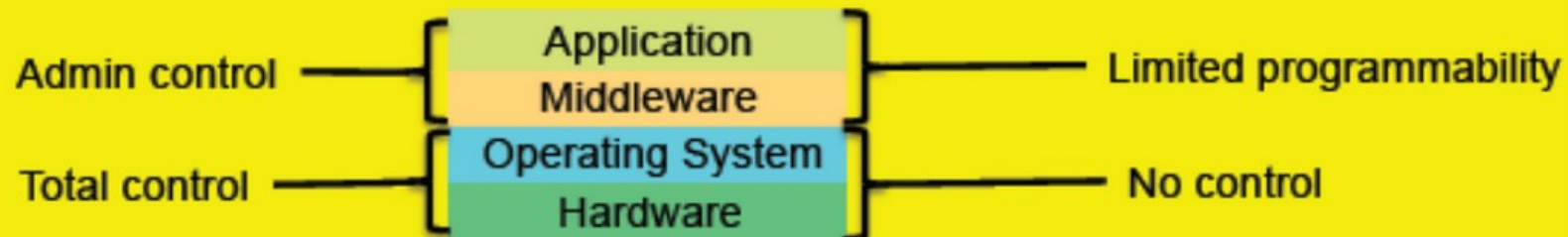
Cloud Provider

Cloud Customer

1 Software as a Service (SaaS)



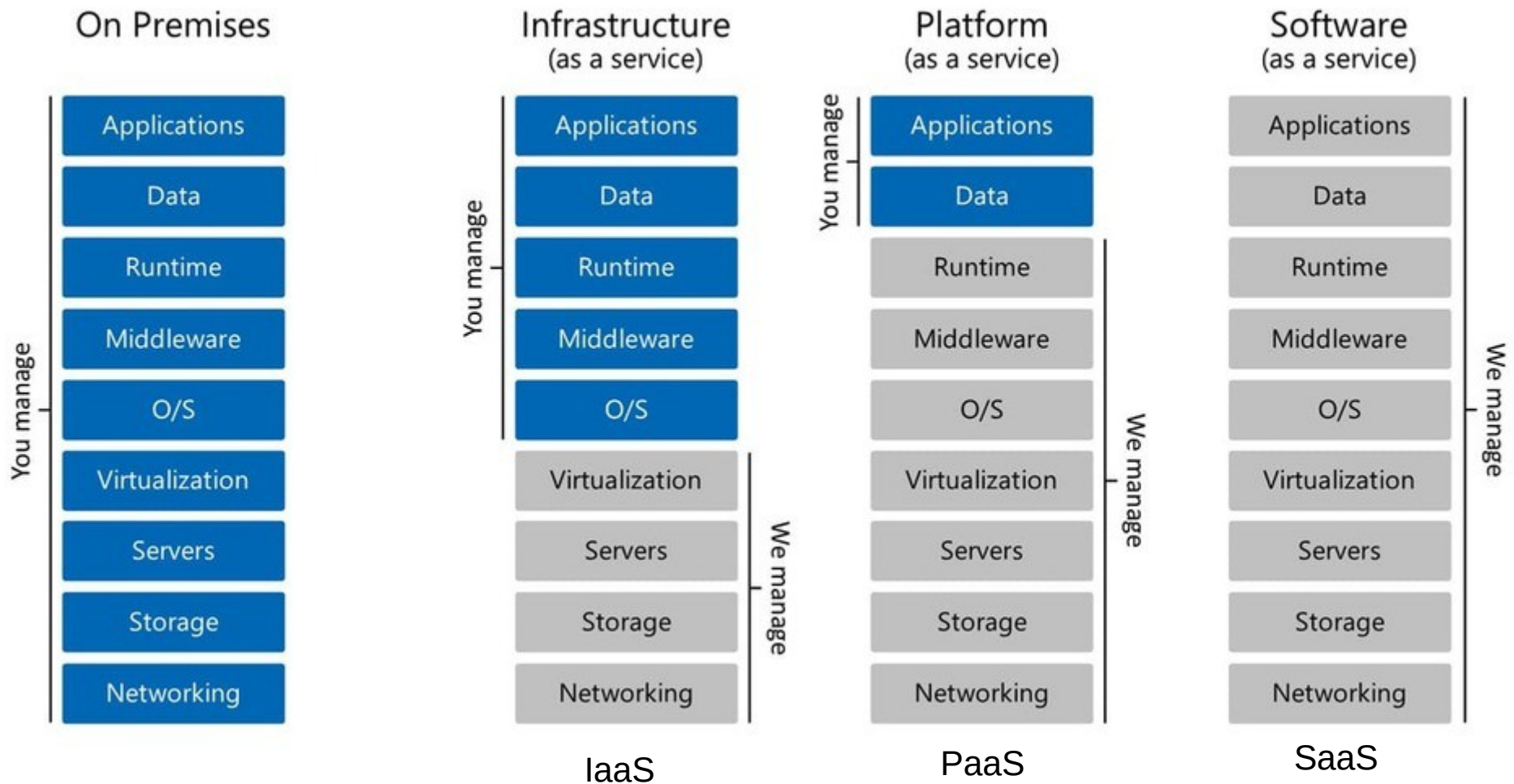
2 Platform as a Service (PaaS)



3 Infrastructure as a Service (IaaS)



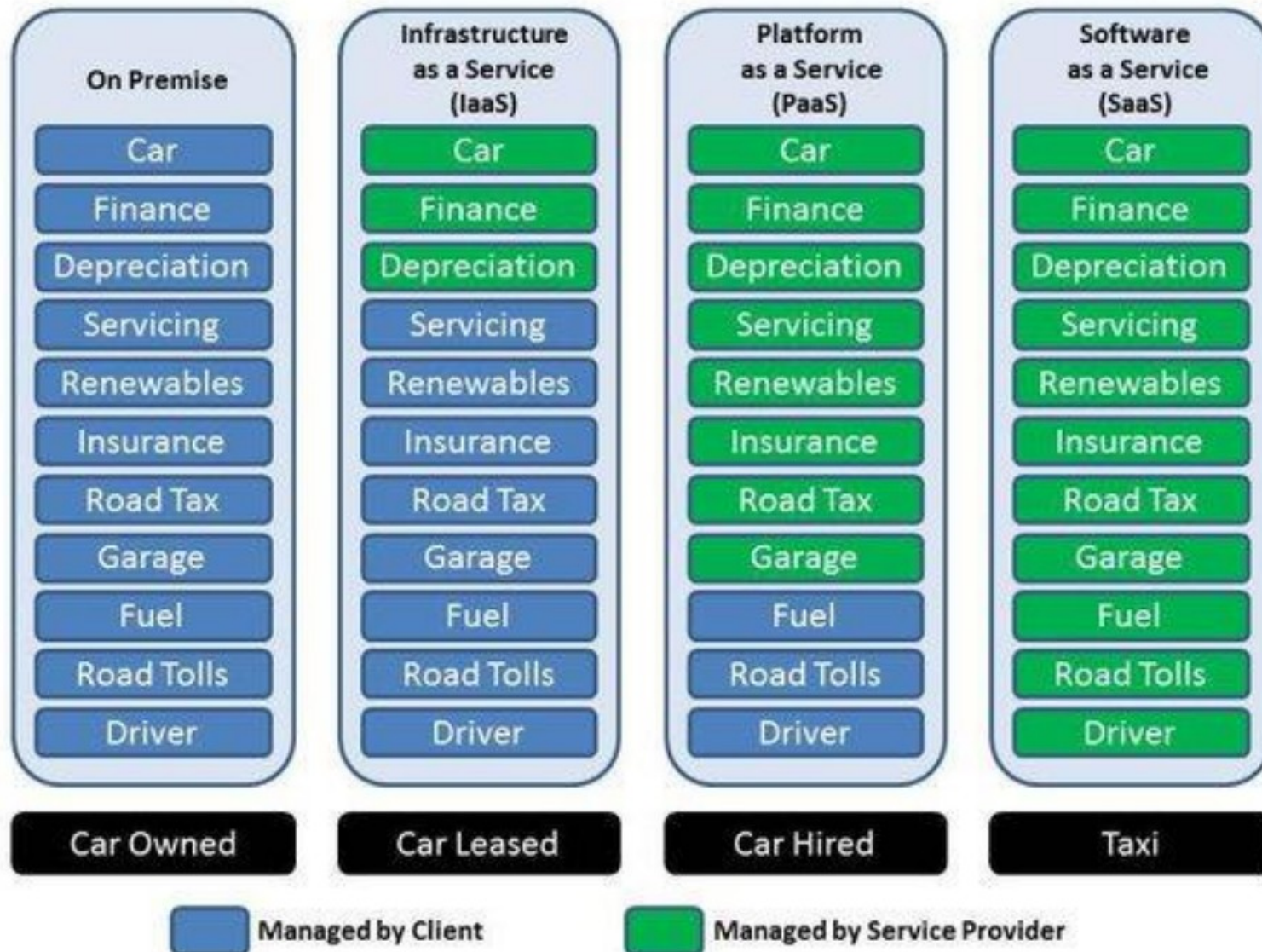
Service paradigms



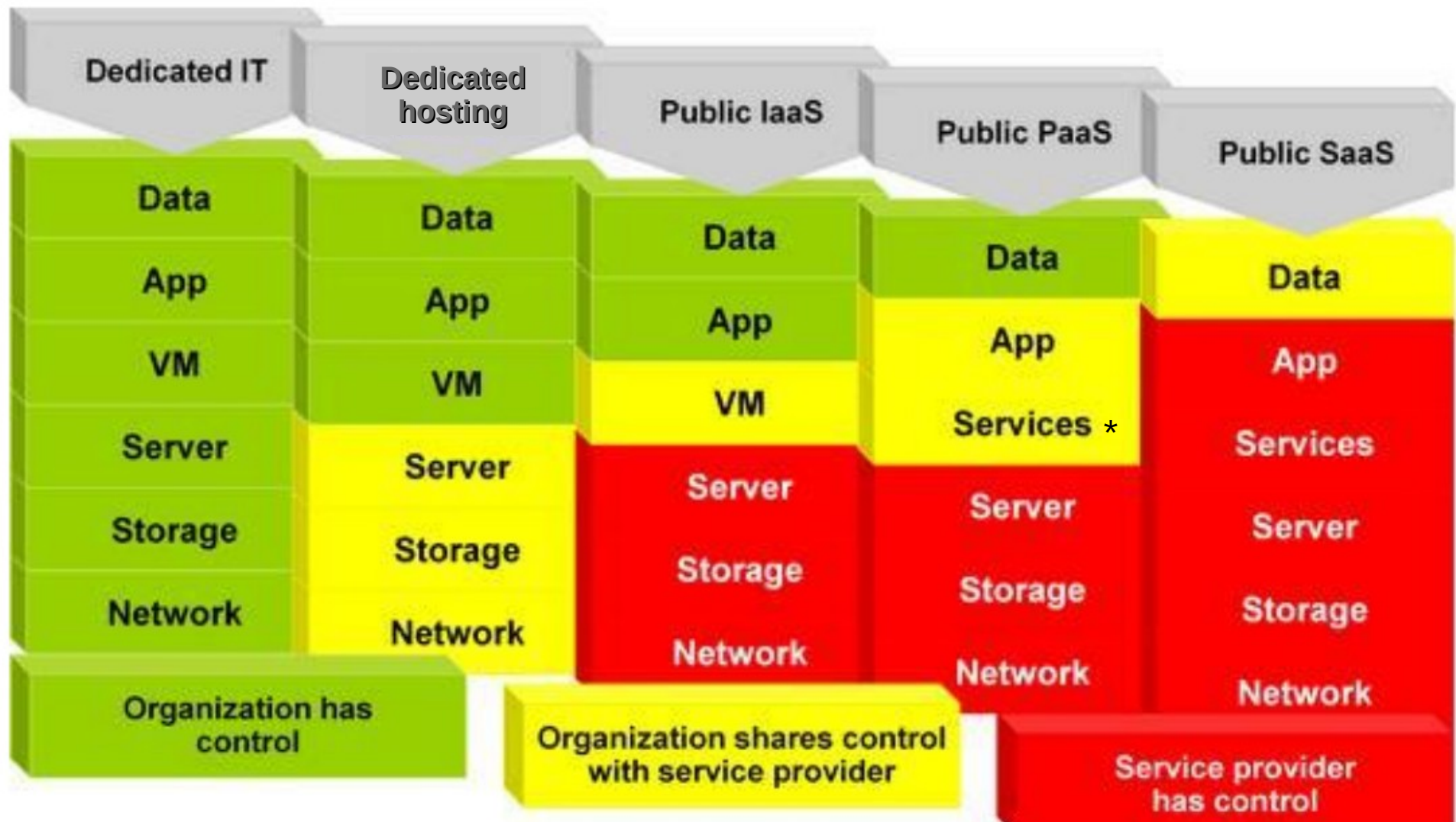
Example in a different context

Car as a Service

alternative



Service paradigms



* Managed Virtual Machines offering to app engine applications some 'service' → a bit more flexibility over app platform, CPU and memory options

laaS (Infrastructure as a Service)

The NIST definition of laaS

“The capability provided to the consumer is **provision processing, storage, networks**, as well as other fundamental computing resources where the consumer is able to **deploy & run arbitrary software**, which can include operating systems and applications.

The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, & deployed applications; and possibly limited control of select networking components (e.g., host firewalls).”

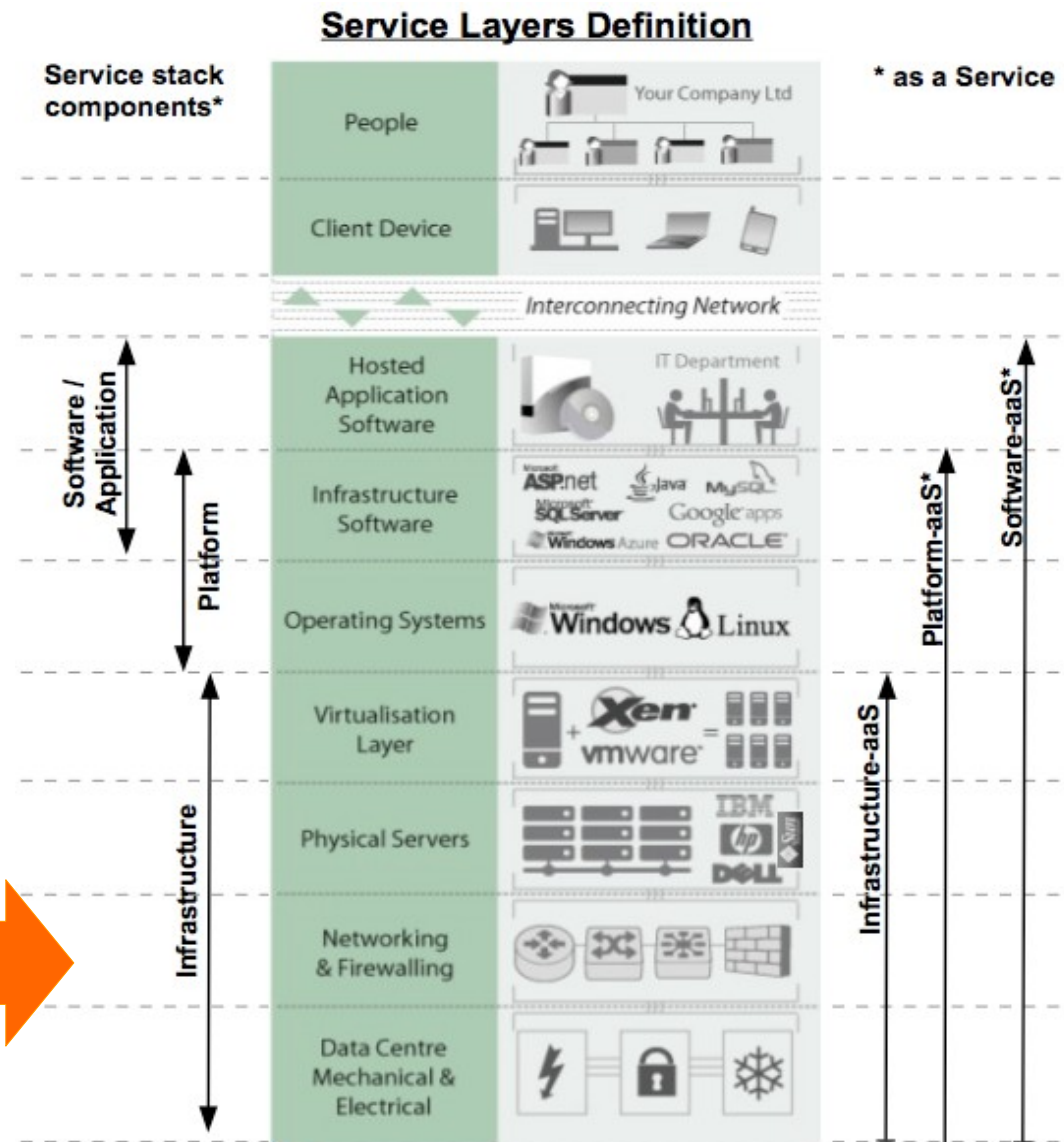
How do we achieve this results of shared resources?
→ **Virtualization, virtual machines**

IaaS (Infrastructure as a Service)

The provider provides “only” the infrastructure part

Advanced outsourcing of all ICT resources: it allows to **rent CPU capabilities, storage, network** to install over them other resources such as **OS, DB and other software**

E.g., Rent of virtual machines as Amazon Elastic Cloud Compute EC2, Amazon Simple Storage Service S3, Google Compute Engine, IBM Blue Cloud, Microsoft Azure IaaS, ...



IaaS Cloud Providers

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Table 1: Data Snapshot: Worldwide IaaS Public Cloud Services Market Share, 2023-2024 ¹⁷⁵

(Millions of U.S. Dollars)

Company	2024 Revenue	2024 Market Share (%)	2023 Revenue	2023 Market Share (%)	2023-2024 Growth (%)
Amazon	64,804.9	37.7	54,664.0	39.0	18.6
Microsoft	41,099.3	23.9	32,205.7	23.0	27.6
Google	15,519.5	9.0	11,458.2	8.2	35.4
Alibaba Group	12,444.9	7.2	11,129.1	7.9	11.8
Huawei	7,082.2	4.1	5,979.7	4.3	18.4
Others	30,803.7	17.9	24,767.5	17.7	24.4
Total	171,754.6	100	140,204.2	100	22.5

Source: Gartner (August 2025)

- In 2024, the top 5 IaaS providers accounted for almost 82% of the market share
- Google saw the highest growth rate of the top five IaaS vendors, for the third year in a row
- Alibaba Group, and Huawei maintained their respective positions year-over-year

PaaS (Platform as a Service)

NIST Definition

“PaaS, or Platform-as-a-Service, is a cloud computing model that provides customers a **complete cloud platform - hardware, software, and infrastructure - for developing, testing, running, and managing applications** without the complexity of building and maintaining the infrastructure typically associated with developing and launching the application(s).”

PaaS offers a **complete development environment that includes tools, libraries and services (APIs), middleware, databases, and operating systems**, simplifying the software development lifecycle

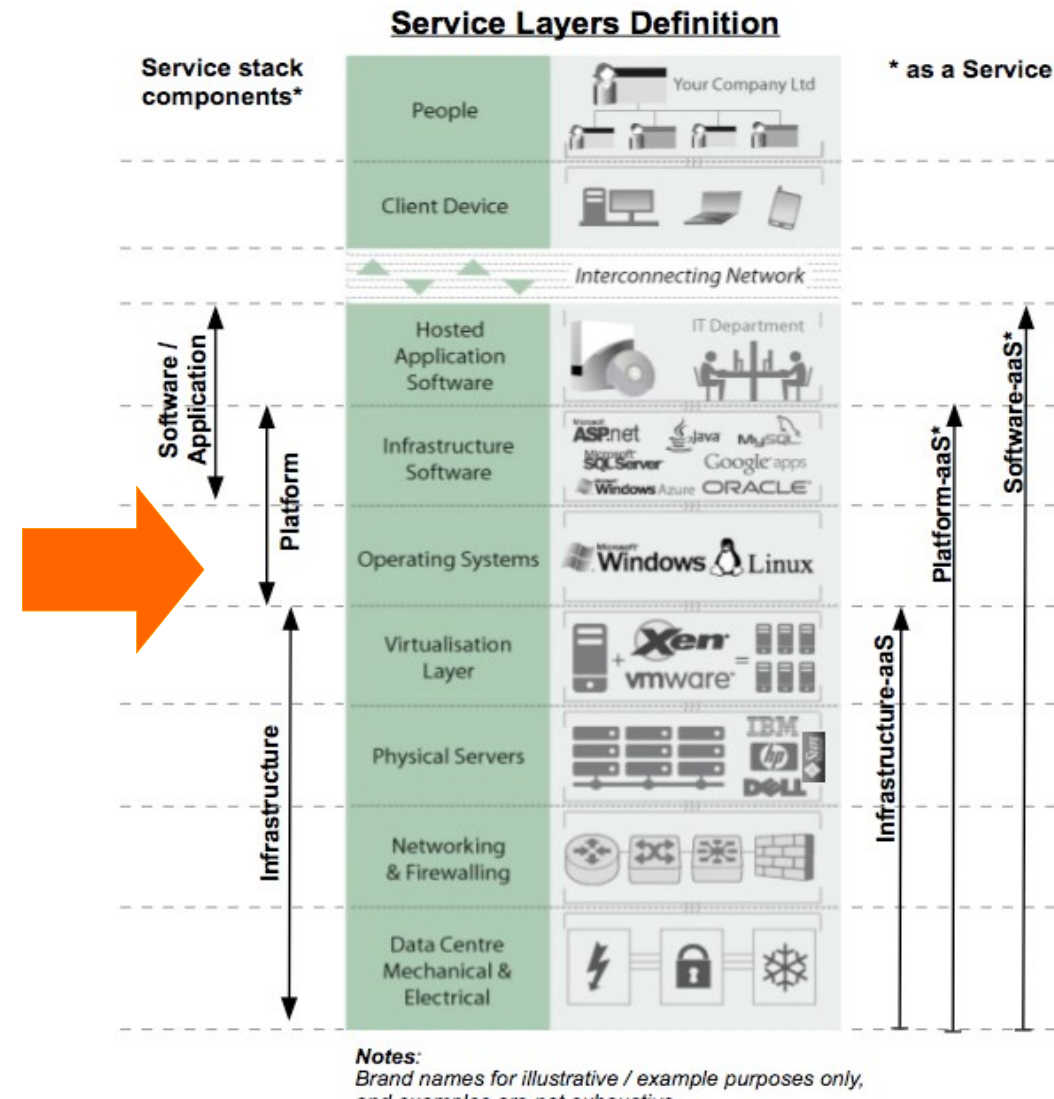
Users can focus on writing code and managing applications, while the PaaS provider handles all the infrastructure resources and maintenance complexities, such as system updates, security, and **scalability**

PaaS (Platform as a Service)

The customer does not have control on the infrastructure, operating systems, or storage, but has control over the developed applications and on their configurations, and often has **support for the automated management of resources** → *monitoring of performance, automated scaling and load balancing*

E.g., Google Cloud App Engine, Amazon Elastic Beanstalk, Amazon API Gateway, Heroku

For example, Amazon API Gateway allows developers to create, publish, maintain, monitor, and secure APIs *at any scale*



PaaS: main providers



SaaS (Software as a Service)

NIST definition

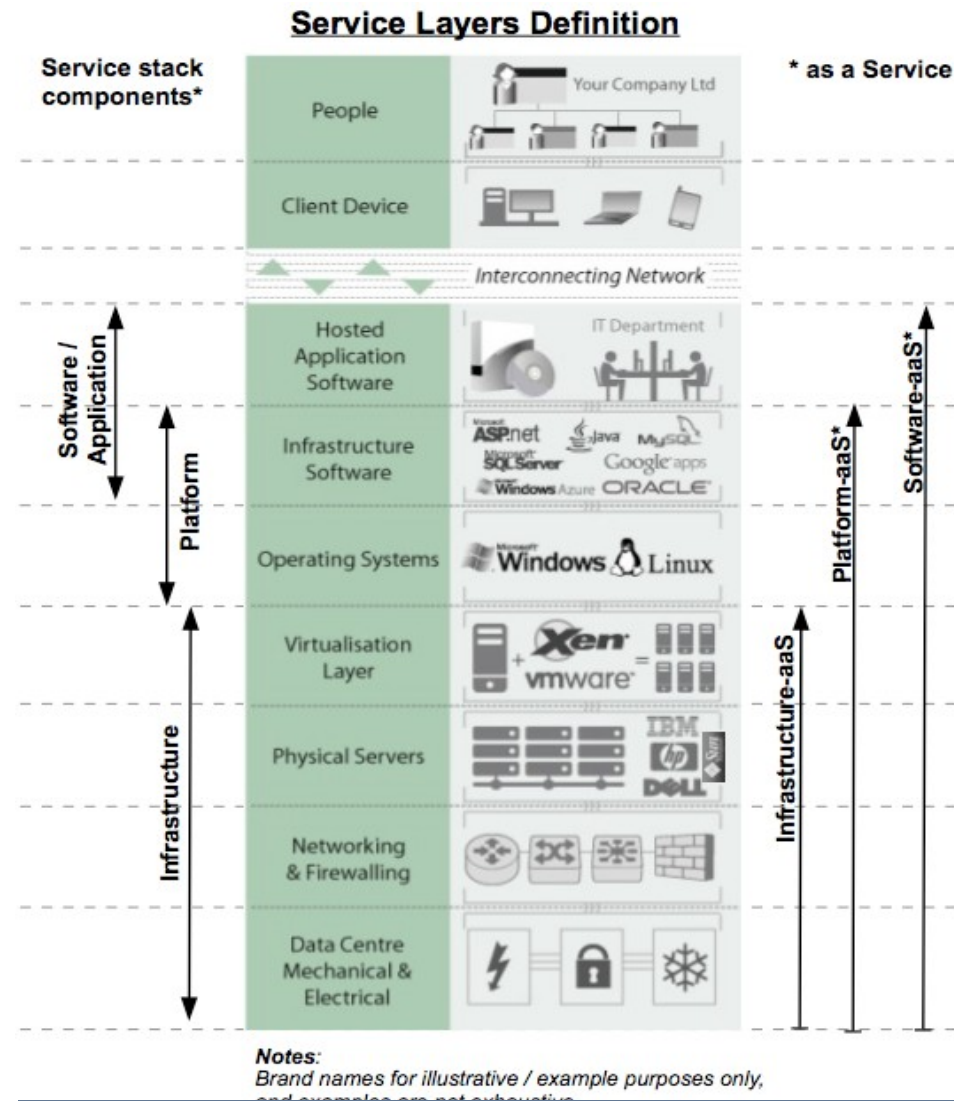
“The capability provided to the consumer is **to use the provider’s applications running on a cloud infrastructure.**

The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface. The consumer **does not manage or control** the underlying cloud infrastructure including network, servers, operating systems, storage, **or even individual application capabilities**, with the possible **exception of limited user-specific application configuration settings.**”

SaaS (Software as a Service)

The provider provides the use of applications on an infrastructure accessible from different devices through a common interface as a browser Web

Examples of SaaS include **tools/applications for shared productivity** (e.g., email Web-based, calendars, Google docs, Google Drive, Microsoft Office 365) as well as **enterprise services** (e.g., deliver human resource software, online ERPs, e-commerce systems, customer relationship management tools, and integrated development environments - IDEs)



SaaS: providers

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IaaS, PaaS, SaaS market share

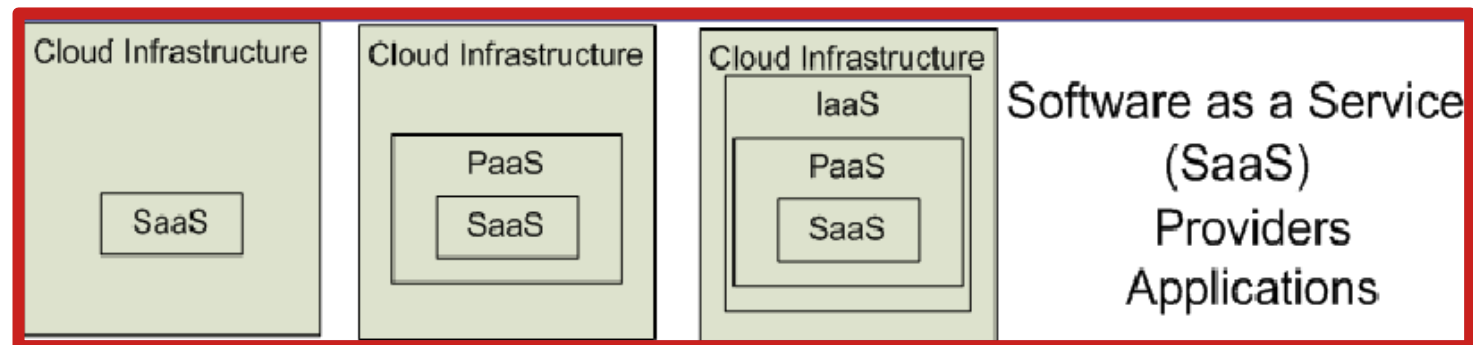
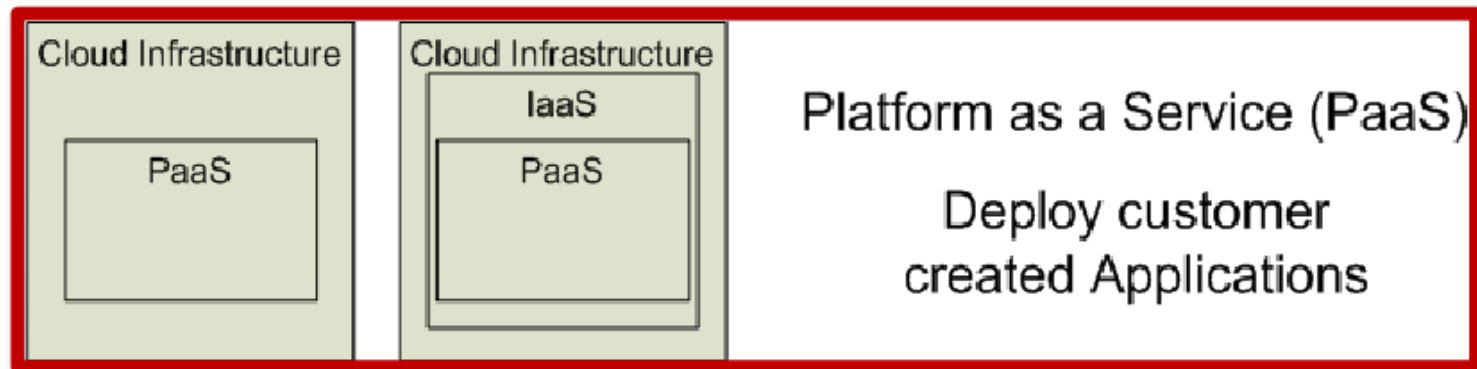
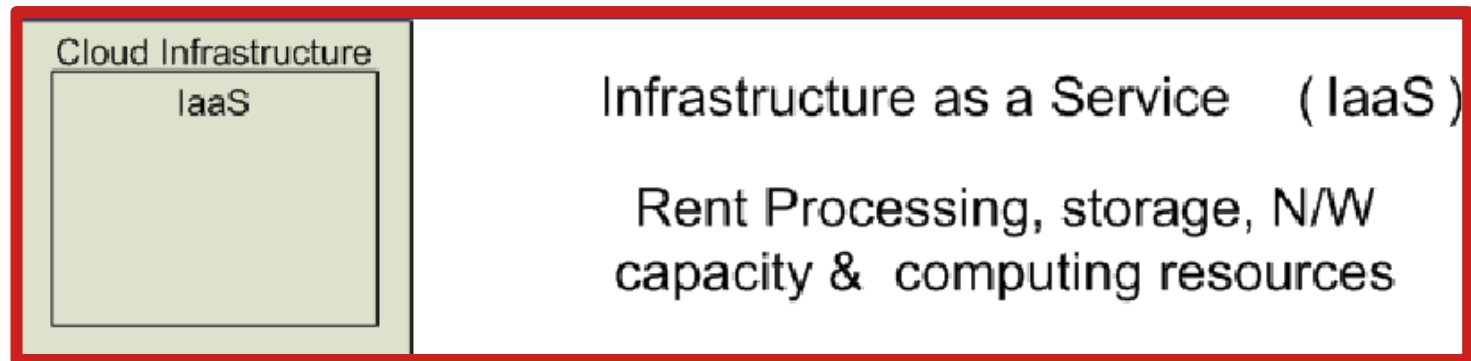
Worldwide Public Cloud Services Revenue and Year-over-Year Growth, 1H 2023 (revenues in US\$ billions)					
Deployment Category	1H23 Revenue	1H23 Market Share	1H22 Revenue	1H22 Market Share	1H23/1H22 Growth
IaaS	\$64.4	20.4%	\$55.1	20.8%	16.9%
PaaS	\$56.8	18.0%	\$44.5	16.8%	27.7%
SaaS – Applications	\$141.2	44.7%	\$121.9	46.0%	15.8%
SaaS – System Infrastructure Software	\$53.2	16.9%	\$43.3	16.4%	22.7%
Total	\$315.5	100%	\$264.8	100%	19.1%
Source: IDC Worldwide Semiannual Public Cloud Services Tracker, 1H 2023, November 30, 2023					

SaaS SIS - System Infrastructure Software - market covers the software solutions that help manage and maintain the underlying infrastructure that supports an organization's applications and data

- Examples of System Infrastructure Software: Firewalls, Information Security, Cryptography, Logging, Location Services, Administration tools, ...

While all markets are **led by a small number of companies**, there continues to be a **healthy long tail of companies** delivering cloud services around the globe, especially in SaaS - Applications

Service models: relationship



Observation: PaaS vs Web hosting

How is PaaS different from traditional Web Hosting (e.g., Aruba, SiteGround)?

1. Typical Web hosting service

Mainly designed to host websites (WordPress, Joomla, e-commerce, etc.).

You buy a preconfigured package that includes disk space, a database (e.g., MySQL), and support for a server-side language (e.g., PHP)

You have a control panel to upload files, manage database, email, etc

The infrastructure is quite rigid: you cannot freely choose languages, frameworks, or runtime versions; you're limited to what the hosting provider offers.

Most of the management is handled by the provider (server patching, basic security, backups, etc.).

Great for small websites, but not suitable for complex or scalable applications

Observation: PaaS vs Web hosting

How is PaaS different from traditional Web Hosting (e.g., Aruba, SiteGround)?

2. Platform as a Service (PaaS) in the Cloud

A higher-level service: it doesn't just host a website, it lets you deploy applications (Web apps, APIs, microservices) using various languages and frameworks (Java, Python, Node.js, .NET, etc.)

The provider manages all the underlying infrastructure (servers, patching, load balancing, scaling), while you focus only on code and application logic

Typical examples: Heroku, Google App Engine, Azure App Service, AWS Elastic Beanstalk

Often includes built-in features like: auto-scaling, monitoring, logging, simplified deployment from Git or CI/CD pipelines.

Suitable for projects that need to grow, since scalability is automated or natively integrated

Observation: PaaS vs Web hosting

To summarise

- Web hosting as Aruba is a very specific and vertical service: it provides a prepackaged environment (web space + PHP + MySQL + control panel) designed mainly for publishing websites.
- Platform as a Service (PaaS) is a broader concept: it gives you a runtime platform for applications, not just websites. You can choose languages, frameworks, libraries, versions, and often benefit from features like automatic scaling
- At most, you could think of Web hosting as a very limited and simplified form of PaaS, because you don't manage the infrastructure (servers, patching, etc.) and only focus on your site's content.
- But in cloud computing terminology, when we say PaaS we mean something much more flexible and general-purpose

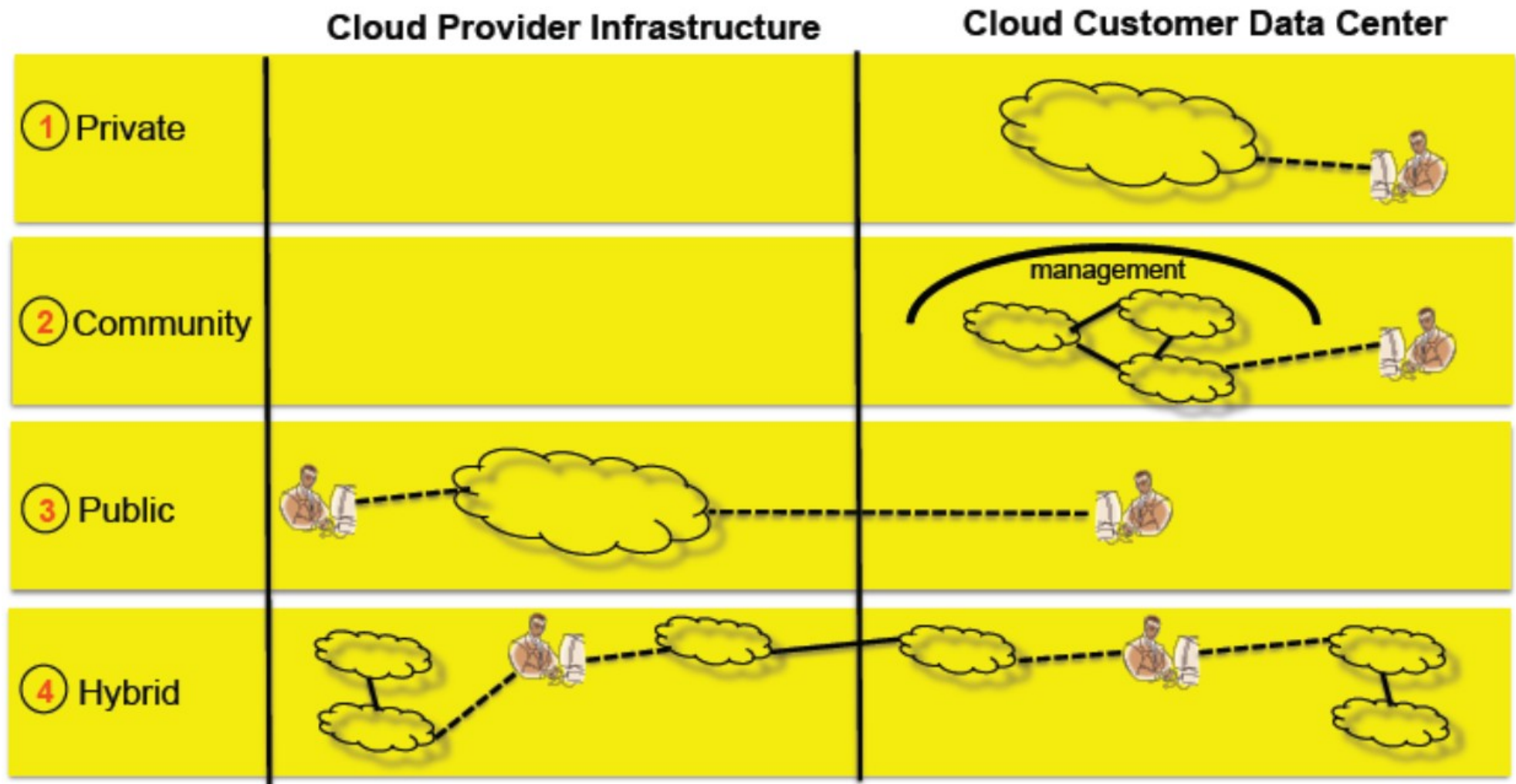
Two main classes of paradigms



Service

Deployment

Deployment paradigms



Deployment models



- **Public cloud.** The cloud infrastructure offers services over the public Internet and is shared by multiple customers. It is owned by an organization (i.e., cloud provider) selling one or more types of cloud services
- **Private cloud.** The cloud infrastructure is operated solely for an organization. It may be managed and hosted on premise or by an external provider (exclusive cloud)
- **Hybrid cloud.** The cloud infrastructure is a composition of two or more clouds (private and public, or all public), that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability, used by the same customer: benefits of multiple deployment models
- **Community cloud.** The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns and/or objectives (e.g., business mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or by a third party and may exist on premise or off premise

Community clouds



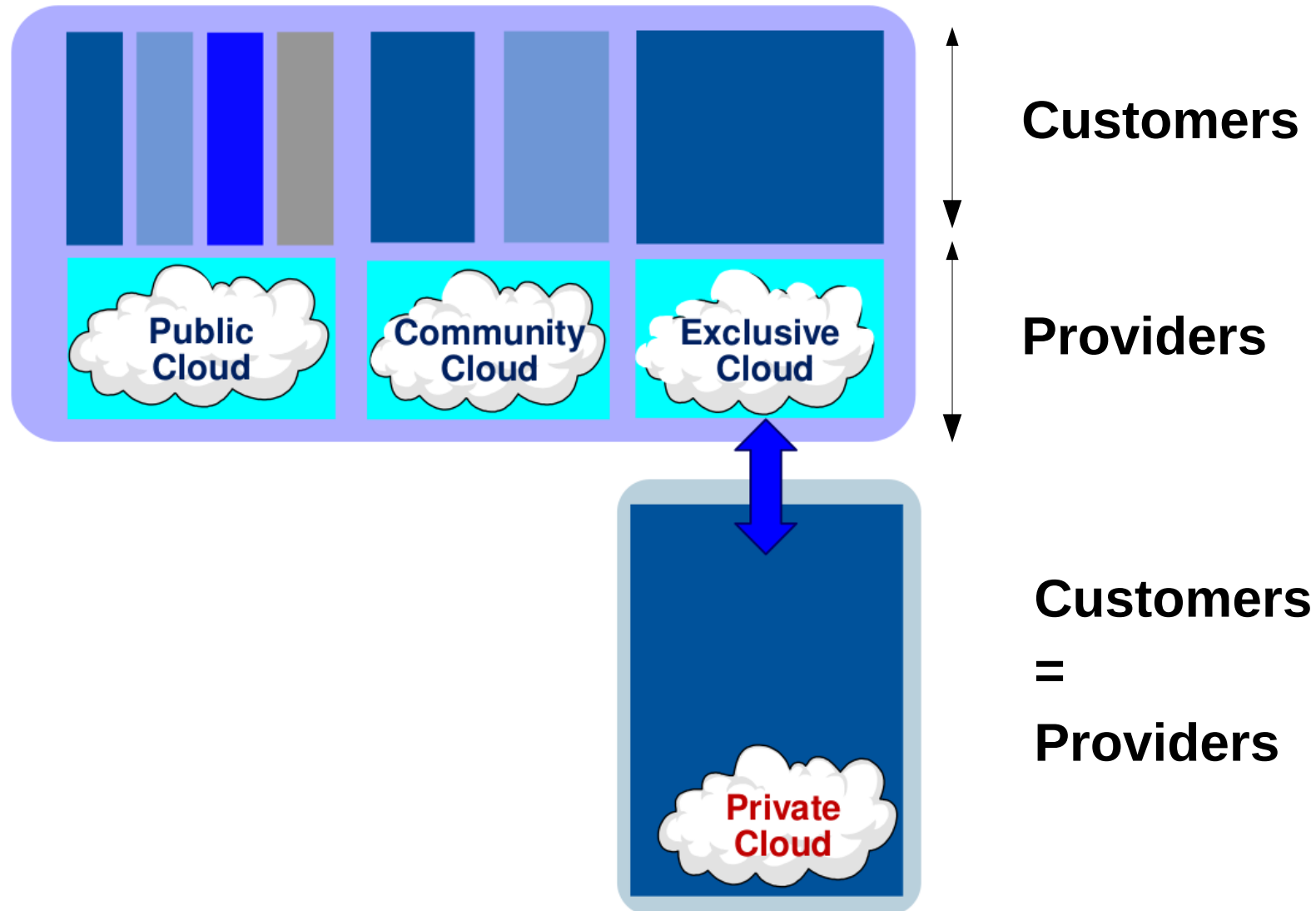
Some examples

- Public Administrations
- Health care systems
- Small-Medium Enterprises
 - Vertical
 - Provider-Clients
- ...

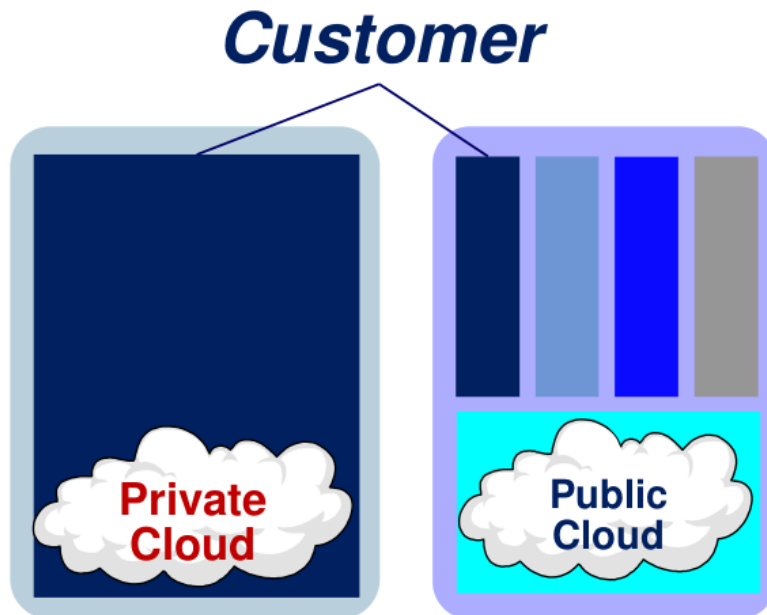
Observation

- The costs are spread over fewer users than a public cloud (but more than a private cloud), so only some of the cost savings potential of cloud computing are realized

Deployment models

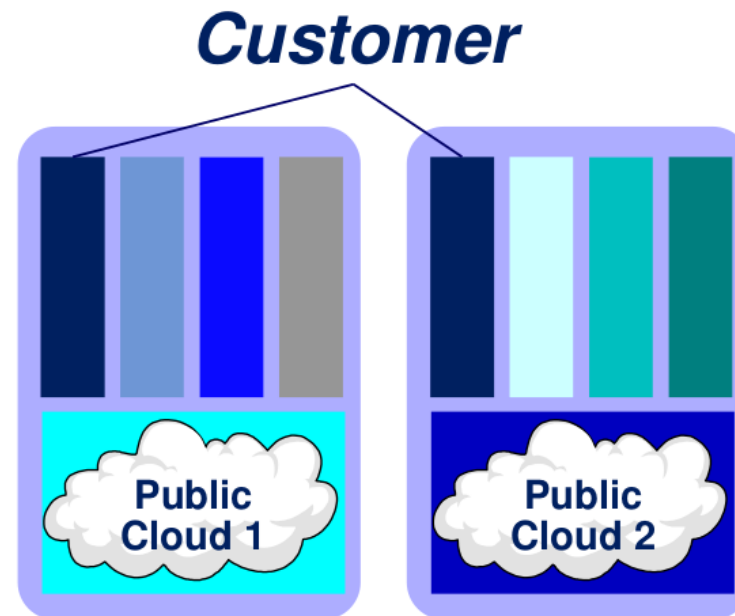


Hybrid solutions



More common definition

Gartner* definition
(*research and consulting company)



An Example: SAP

How is the service paradigm of SAP ERP?

SAP can run on three main types of infrastructures

On-premise (Local Infrastructure):

The company manages and hosts the entire physical infrastructure and runs SAP in its own data centers: approach preferred by companies with very strict security requirements or those that want complete control over the infrastructure

Public Cloud:

SAP ERP can run on public cloud infrastructures provided by vendors like Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), and others. SAP offers a cloud-native solution called **SAP HANA Cloud**, which is an optimized and scalable version of their ERP, designed to run on public clouds.

Hybrid Cloud:

A hybrid approach combines private cloud with public cloud: some resources kept on-premise, while other parts of the ERP in the cloud. This allows companies to balance security and flexibility, keeping sensitive data on local infrastructure while using the cloud to easily scale resources when needed.

SAP HANA Cloud

SAP HANA Cloud utilizes both its own data centers and those of third-party cloud providers

SAP Proprietary Data Centers:

SAP manages a network of its own data centers in various locations around the world. SAP data centers are particularly suited for customers requiring a high level of control, customization, or regulatory compliance.

Public Cloud Partners (AWS, Azure, Google GCP):

SAP leverages the infrastructure of public cloud partners to offer scalable, flexible, and geographically distributed services, supporting a wide range of customer needs.

This approach allows SAP to take advantage of elastic scalability, global availability zones, and advanced services offered by leading public cloud providers while maintaining control over the SAP management and application layer.

Hybrid Approach:

Parts of the infrastructure can remain on-premise (or in private data centers), while other parts can be hosted in the public cloud or in SAP data centers. This allows companies to choose the solution best suited to their specific needs in terms of security, performance, and compliance.

Cloud Providers

Cloud providers

Only a handful of major IT players can really build a similar massive infrastructure

amazon.com



Microsoft

Windows Azure Platform



Alibaba Cloud

Google

Google app engine

IBM



Worldwide telco operators are accessing the cloud market



Recently, other big companies



Specialized Cloud Providers

- **Akamai** → Content delivery (market leader) and security services
- **Cloudflare** → Security (35% DNS, API and SaaS protection, DDoS protection, SSL communications) and Content Delivery



The big players [Gartner, 2013 and 2014]



Magic Quadrant for Strategic Cloud Platform Services: IaaS + PaaS + transformation services



The distance augments [Gartner, 2018]



The big players [Gartner, 2023 and 2024]



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Amazon AWS vs. MS Azure vs. Google cloud



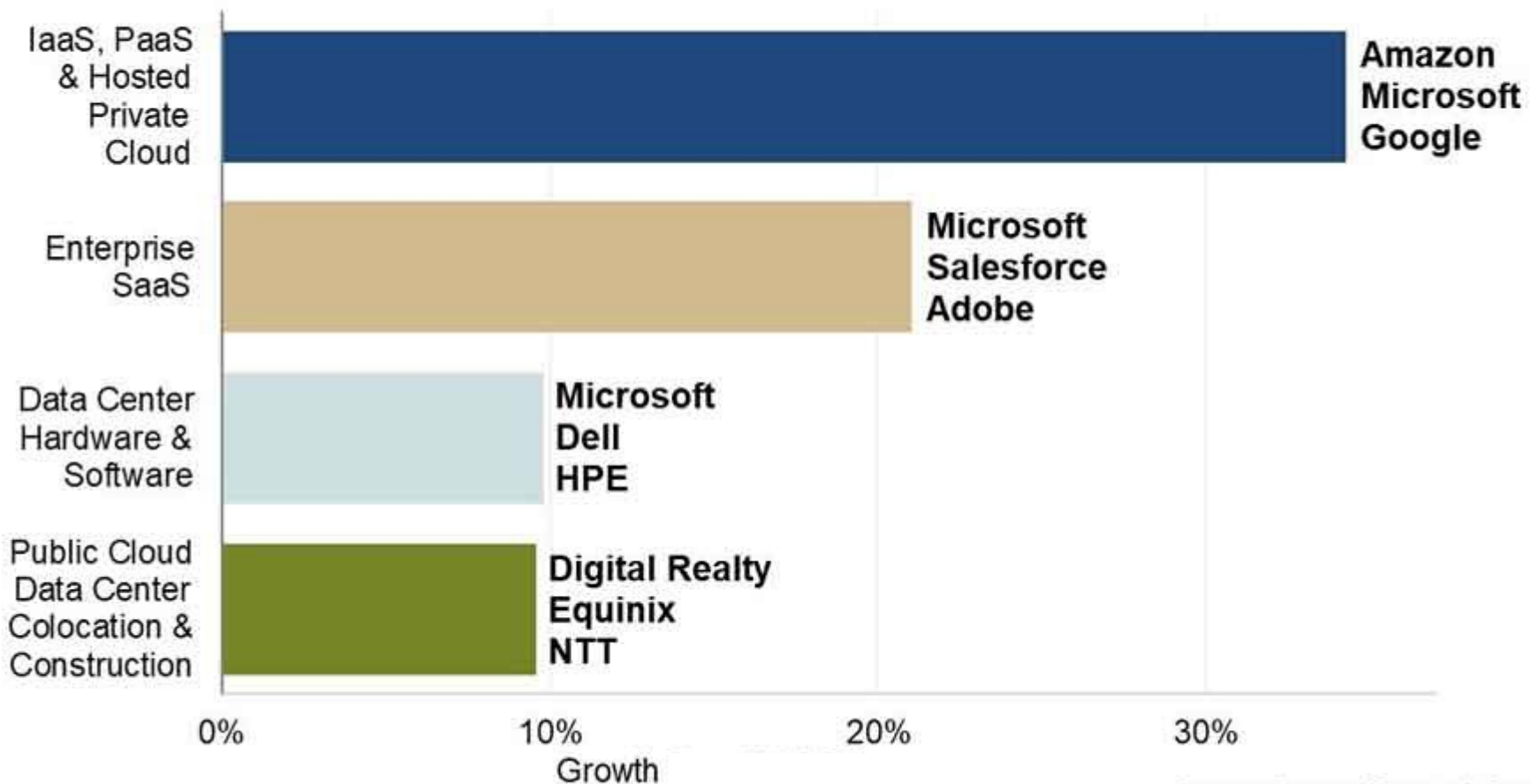
- **Amazon AWS leads the pack.** Who would have guessed that 15 years ago, an online book seller would lead the pack in offering IT infrastructure? Now it's also a matter of “**critical mass**”
- **Microsoft's Azure is coming close** to AWS in terms of offering especially thanks to the Office 365 ecosystem. Still even too complex contracts
- **Google cloud** has some boss technologies and he learned to engage with midmarket and enterprise customers”. Its main business has traditionally been more in the PaaS-SaaS context than in the IaaS arena

Today, it is difficult to classify one cloud provider in just one of the three main service classes:

IaaS, PaaS, SaaS

Segment leaders

Cloud Market Growth & Segment Leaders

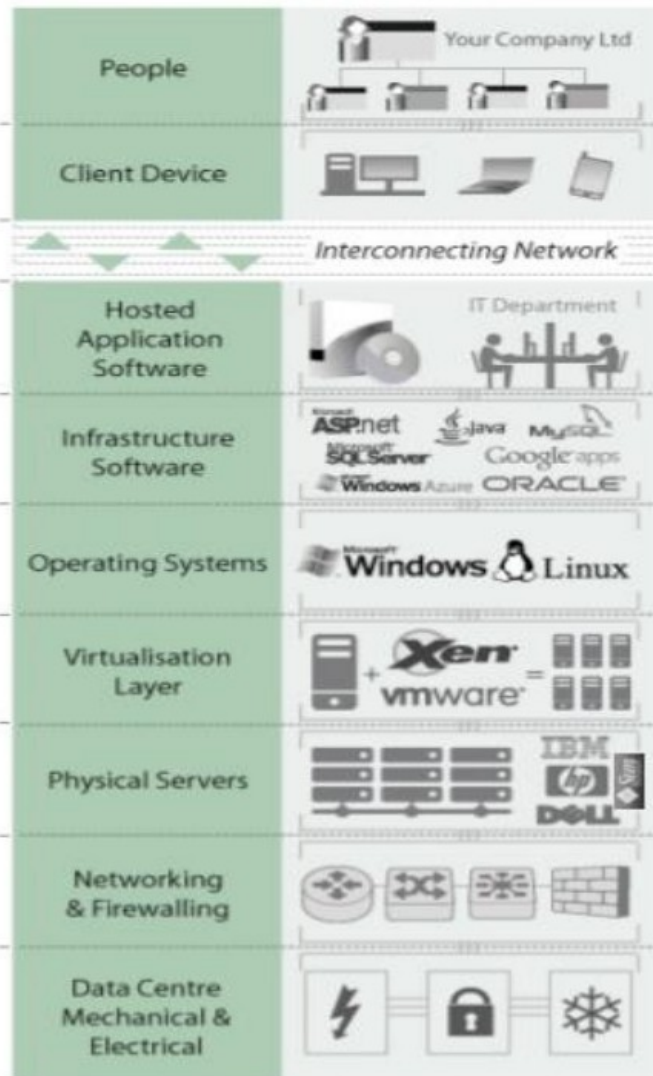


Source: Synergy Research Group

More than 3 paradigms

Service Layers Definition

Service stack components*



* as a Service

The need to support cloud services integration and multiple clients require novel paradigms

- iPaaS
- RaaS
- FaaS*

*Function as a Service
(we will see)

Notes:
Brand names for illustrative / example purposes only,
and examples are not exhaustive.

IpaaS, RaaS



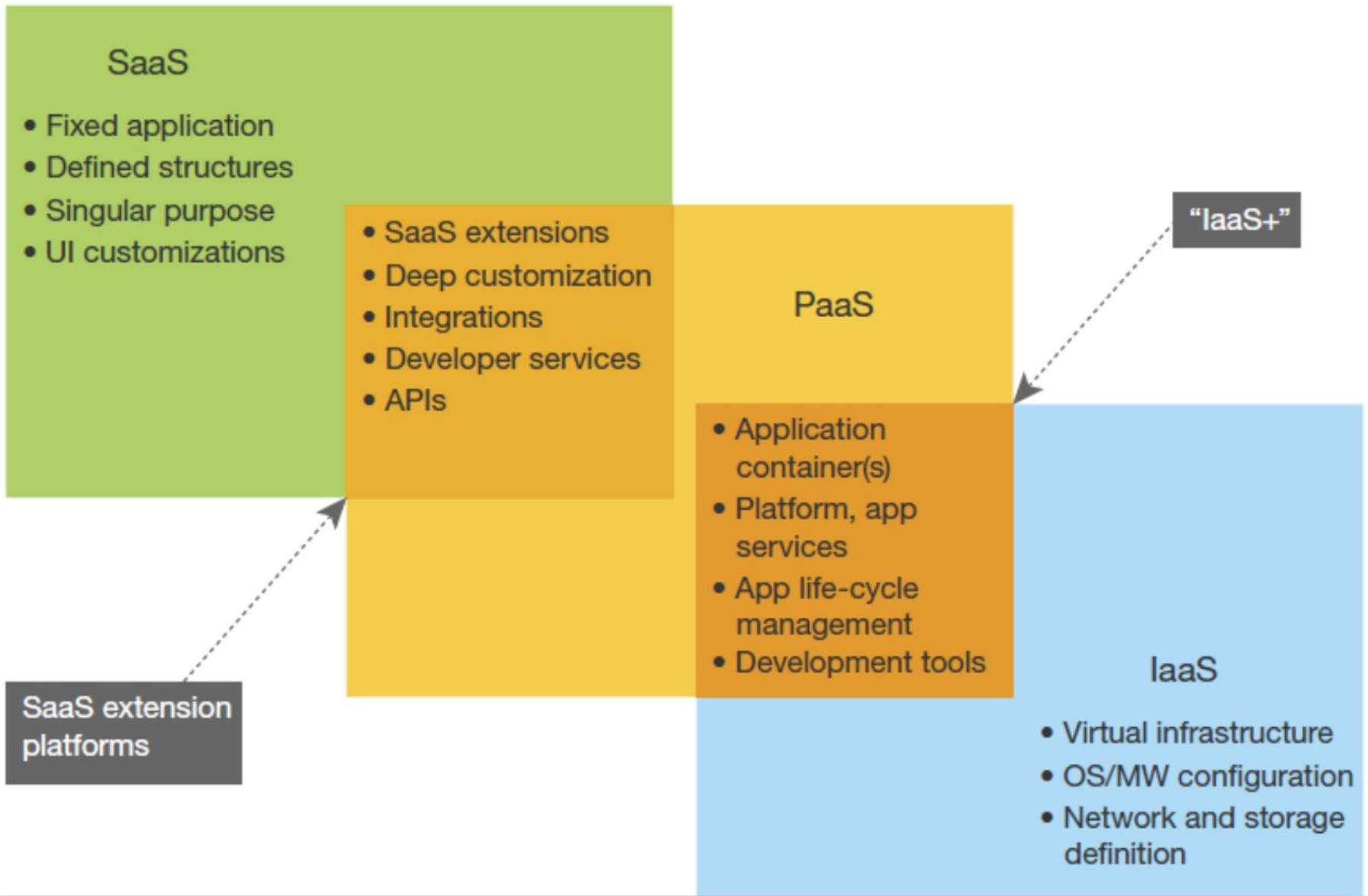
iPaaS - integration Platform as a Service

- Suite of cloud services enabling customers to develop, execute and manage integration flows between different applications
- Integration for cloud-based software applications or for cloud to on-premises applications

RaaS - Recovery as a Service

- Service used to protect data or applications from natural / human disasters or service disruption at one location by enabling full recovery
- Differs from cloud-based backup services by protecting data and providing standby computing capacity on demand to facilitate more rapid application recovery
- Delivered in a cloud-computing model: recovery resources paid when they are used → more efficient than traditional disaster recovery site with resources always running
 - **To-cloud RaaS:** source application in users primary private datacenter
 - **In-cloud RaaS:** both the source and recovery sites are in the cloud
 - **From-cloud RaaS:** primary or production application or data is in the cloud and the backup or recovery target site is a private datacenter

Disappearing boundaries



IaaS+

- Historical IaaS vendors (e.g., Amazon, Rackspace) are building services that deliver the abstraction and hosted middleware benefits that were often associated with PaaS
- **Examples of IaaS+**
 - Managed load balancers (e.g., AWS ELB) → still infrastructure, but with added intelligence and automation to help manage it
 - AWS CloudFormation: create templates to describe AWS resources and dependencies to help manage them → management/orchestration service that sits on top of IaaS
 - Elastic MapReduce (process vast amount of data): technically PaaS, but often considered IaaS+ because it orchestrates Hadoop/Spark clusters on EC2/S3 (i.e., infrastructure-based but with added abstraction).

PaaS+IaaS

- Traditional PaaS vendors are reaching down to the virtual infrastructure
- Some PaaS providers now offer both highly abstracted interface for developers and the ability to configure underlying middleware, database, and some IaaS resource
- **Examples**
 - Heroku, one of Salesforce's cloud platforms, runs on AWS, and the vendor encourages developers to mix Heroku and AWS services

SaaS+PaaS



SaaS+PaaS

- SaaS vendors are pushing into platforms with extensibility features
- **Examples**
 - Salesforce: core is SaaS (CRM - (Customer Relationship Management), but with Force.com and Lightning Platform you can build custom applications inside the Salesforce ecosystem → that's PaaS inside SaaS.
 - Microsoft Office 365: SaaS suite, but exposes APIs, Graph API, and add-in development frameworks → allowing developers to extend functionality.